Nischal Kharel

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Education

University of Pittsburgh Expected Graduation: May 2027

BS in Computer Science, Minor in Information Science *GPA: 3.75/4.0*

- Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Software Engineering, Database Management, Machine Learning, Data Engineering
- Undergraduate Teaching Assistant for Data Structures and Algorithms
- Undergraduate Research Assistant
- High School Valedictorian

Experience

Software Engineer Internship June 2026 – August 2026

BNY *Pittsburgh, PA*

- Streamlined legacy mainframe operations by designing and deploying Java Spring Boot agentic workflows using LangChain4j, successfully replacing manual processes with automated, event-driven data pipelines.
- Architected Kafka consumer workflows to intake, transform, and normalize high-throughput real-time data streaming across microservices.
- Engineered intelligent decision frameworks using Camunda (BPMN/DMN) and external API integrations to parse, validate, and contextualize complex incoming data payloads.

Software Engineer Co-op January 2026 – April 2026

BNY *Pittsburgh, PA*

- Architected and deployed 4 RESTful API endpoints within a Spring Boot microservice to expose enterprise financial data to internal AI agents, enabling seamless data retrieval for the operations team.
- Modernized service-to-service communication by replacing legacy RestTemplate dependencies with reactive WebClient, improving maintainability and aligning with modern Spring design patterns.
- Resolved 100% of LLM tool invocation failures across 40+ MCP tools by diagnosing and fixing a critical type-mapping bug, implementing a type-aware schema conversion that restored support for non-string parameters.

Software Engineer Co-Op May 2025 – August 2025

BNY *Pittsburgh, PA*

- Streamlined account activation by orchestrating an automated pipeline: an email listener forwards client files to an AI Agent for extraction and transformation, feeding the transformed data directly into the BNY data warehouse.
- Engineered intricate prompts for multiple AI Agents to accurately extract data from dynamic Smartform OCR Documents, achieving 98% average accuracy and returning data in JSON format.
- Optimized document processing time by designing and implementing a parallel Agent execution framework using Python's ThreadPoolExecutor and internal API calls, significantly reducing extraction latency.

Projects

GraphRAG Code Expertise Finder | *Python, LangGraph, Neo4j, PostgreSQL* Sept 2026 – Present

- Co-developed a CGI-sponsored GraphRAG chat assistant that answers questions like "Who worked on Jira authentication?" about a GitHub repository, citing the exact commits and contributors behind each answer and drawing them as a graph.
- Engineered its LangGraph question-answering pipeline: an LLM classifies each query, the pipeline retrieves matching commits with their authors and files from Neo4j, and a second LLM writes the cited answer.
- Implemented semantic search across Git history by indexing LLM commit summaries in a Neo4j vector index (Cypher 25), outperforming raw message matching while embedding 180+ commits for under $0.001.
- Ensured every citation shown links to a real commit or contributor by validating LLM references against retrieved graph nodes, scrubbing emails before external model calls, and authoring 340+ unit and live integration tests.

Skills

Languages: Java, Python, HTML, CSS, JavaScript, C
Frameworks: LangChain, LangChain4j, React, Spring Boot, LangGraph, Flask
Libraries: pandas, NumPy, Matplotlib
Developer Tools: Camunda, Git, VS Code, Visual Studio
Databases: MySQL, Neo4j, Snowflake, PostgreSQL
AI: RAG, AzureOpenAI, MCP, GraphRAG